An attendee of Peebles's talk, Ken Turner, went to visit the Arecibo radio telescope, and on the flight back, had a chat with fellow astronomer Bernard Burke about how cool it would be to detect this Big Bang radiation. After arriving back at the office, Burke got a phone call from Penzias about some unrelated work, and happened to mention the airplane chatter.[fn9] At which point, I can only assume Penzias had to have a bit of a sit-down, because now he knew that he and Wilson had just become the first human beings to *see the actual Big Bang*. He took a couple days, talked with his colleague, and then phoned up Robert Dicke, who immediately turned around to Peebles and Wilkinson and said, "We've been scooped."

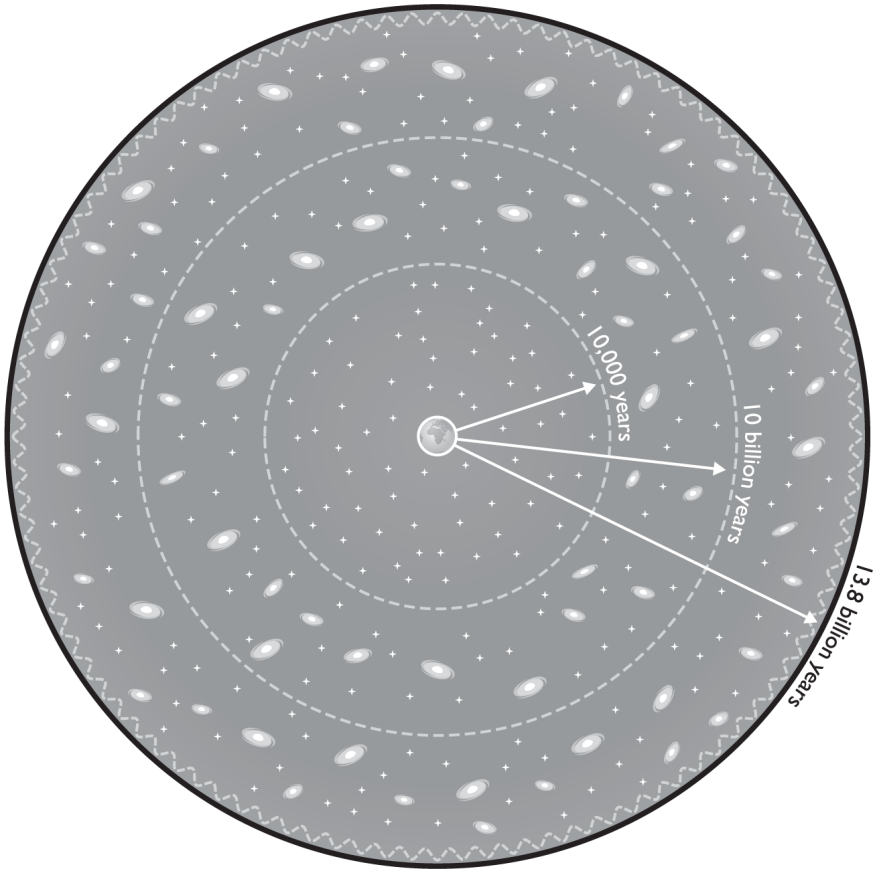


Figure 3: A cartoon map of the observable universe. At different distances from our vantage point on Earth, we can see different epochs in the past. The lookback time (the number of years before today) is labeled for each sphere around us in this diagram. The farthest we can see, even in principle, corresponds to a distance from Earth from which light leaving that point at the very beginning of the universe would be reaching us right now. This defines a sphere around us known as the *observable universe*.